Chapter 04 : Change of Axes

Ques 01 : Transform to parallel axes through the new origin of the equations.

(a) Origin, (1, -2), $2x^2 + y^2 - 4x + 4y = 0$

Putting $x = x_1 + 1$ and $y = y_1 - 2$, we get –

$$2(x_1 + 1)^2 + (y_1 - 2)^2 - 4(x_1 + 1) + 4(y_1 - 2) = 0$$

0r, $2x_1^2 + 4x_1 + 2 + y_1^2 - 4y_1 + 4 - 4x_1 - 4 + 4y_1 - 8 = 0$

or, $2x_1^2 - 6 + y_1^2 = 0$

Removing suffixes we get-

$2x^2 + y^2 = 6$

(b) Origin, (3, 1), $x^2 - 6x + 2y^2 + 7 = 0$

Putting $x = x_1 + 3$ and $y = y_1 - 1$, we get –

$$(x_1 + 3)^2 - 6(x_1 + 3) + 2(y_1 - 1)^2 + 7 = 0$$

or, $x_1^2 + 6x_1 + 9 - 6x_1 - 18 + 2y_1^2 - 4y_1 + 2 + 7 = 0$

or, $x_1^2 + 2y_1^2 - 4y_1 = 0$

Removing suffixes we get-

$x^2 + 2y^2 - 4y = 0$

Ques 02 : Transform to axes inclined at 45° to the original axes the equations.

(i) $x^2 - y^2 = a^2$

Putting, $x = x_1 cos45° - y_1 sin45° = \frac{x_1 - y_1}{\sqrt{2}}$ and $y = x_1 sin45° + y_1 cos45° = \frac{x_1 + y_1}{\sqrt{2}}$, we get-

$$\left(\frac{x_1 - y_1}{\sqrt{2}}\right)^2 - \left(\frac{x_1 + y_1}{\sqrt{2}}\right)^2 = a^2$$

Or, $\frac{(x_1 - y_1)^2}{2} - \frac{(x_1 + y_1)^2}{2} = a^2$

or, $x_1^2 - 2x_1y_1 + y_1^2 - x_1^2 - 2x_1y_1 - y_1^2 = a^2$

or, $-4x_1y_1 = 2a^2$

$\therefore\ -2x_1y_1 = a^2$

Removing the suffixes, we get-
$-2xy = a^2$

(ii) $\boldsymbol{x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0}$

Putting, $x = x_1 cos45° - y_1 sin45° = \frac{x_1 - y_1}{\sqrt{2}}$ and $y = x_1 sin45° + y_1 cos45° = \frac{x_1 + y_1}{\sqrt{2}}$, we get-

$\left(\frac{x_1 - y_1}{\sqrt{2}}\right)^2 - \left(\frac{x_1 + y_1}{\sqrt{2}}\right)^2 - 2\sqrt{2}\left(\frac{x_1 - y_1}{\sqrt{2}}\right) - 10\sqrt{2}\left(\frac{x_1 + y_1}{\sqrt{2}}\right) + 2 = 0$

Or, $(x_1 - y_1)^2 - (x_1 + y_1)^2 - 4(x_1 - y_1) - 20(x_1 + y_1) + 4 = 0$

Or, $x_1^2 - 2x_1y_1 + y_1^2 - x_1^2 - 2x_1y_1 - y_1^2 - 4x_1 + 4y_1 - 20x_1 - 20y_1 + 4 = 0$

Or, $-4x_1y_1 - 24x_1 - 16y_1 = -4$

$\therefore x_1y_1 + 6x_1 + 4y_1 = 1$ [dividing both sides by -4]

Removing suffixes, we get-
$xy + 6x + 4y = 1$

Ques 03 : Remove the first degree terms in each of the following equations.

(a) $\boldsymbol{3x^2 - 4y_2 - 6x - 8y - 10 = 0}$

Given that,
$3x^2 - 4y_2 - 6x - 8y - 10 = 0$
or, $3(x^2 - 2x + 1) - 4(y^2 + 2y + 1) - 9 = 0$
or, $3(x - 1)^2 - 4(y + 1)^2 = 9$ .....(1)

Putting $x - 1 = x_1$ and $y + 1 = y_1$ in eq. (1), we get-
$3x_1^2 - 4y_1^2 = 9$

Removing the suffixes-
$3x^2 - 4y^2 = 9$

(b) $\boldsymbol{2x^2 + 5y^2 - 12x + 10y - 7 = 0}$

Given that,
$2x^2 + 5y^2 - 12x + 10y - 7 = 0$
or, $2(x^2 - 6x + 3^2) + 5(y^2 + 2y + 1) - 7 = 23$
or, $2(x - 3)^2 + 5(y + 1)^2 = 30$

Putting $x-3=x_1$ and $y+1=y_1, we\ get -$
$2x_1^2+5y_1^2=30$

Removing suffixes, we get-
$2x^2+5y^2=30$

(b) $\mathbf{3x^2-4y^2+6x+24y-135=0}$

Given that,
$3x^2-4y^2+6x+24y-135=0$
or, $3(x^2+2x+1)-4(y^2-6y+3^2)-135=-33$
or, $3(x+1)^2-4(y-3)^2=102$

Putting $x+1=x_1$ and $y-3=y_1$, we get -
$3x_1^2-4y_1^2=102$

Removing suffixes -
$3x^2-4y^2=102$

Ques 04 : Prove that the value of g^2+f^2 in the equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ remains uneffected by orthogonal transformation without change of origin.

Putting $x=x_1cos\theta-y_1sin\theta$ and $y=x_1sin\theta+y_1cos\theta$ in the given equation, we get –

$a(x_1cos\theta-y_1sin\theta)^2+2h(x_1cos\theta-y_1sin\theta)(x_1sin\theta+y_1cos\theta)+b(x_1sin\theta+y_1cos\theta)^2+2g(x_1cos\theta-y_1sin\theta)+2f(x_1sin\theta+y_1cos\theta)+c=0$

Or, $a(x_1cos\theta-y_1sin\theta)^2+2h(x_1cos\theta-y_1sin\theta)(x_1sin\theta+y_1cos\theta)+b(x_1sin\theta+y_1cos\theta)^2+2gx_1cos\theta-2gy_1sin\theta+2fx_1sin\theta+2fy_1cos\theta+c=0$

Or, $a(x_1cos\theta-y_1sin\theta)^2+2h(x_1cos\theta-y_1sin\theta)(x_1sin\theta+y_1cos\theta)+b(x_1sin\theta+y_1cos\theta)^2+2x_1(gcos\theta+fsin\theta)+2y_1(fcos\theta-gsin\theta)+c=0$

Here,
$(gcos\theta+fsin\theta)^2+(fcos\theta-gsin\theta)^2$
$=g^2\cos^2\theta+2gfcos\theta sin\theta+f^2\sin^2\theta+f^2\cos^2\theta-2gfcos\theta sin\theta+g^2\sin^2\theta$
$=g^2(\cos^2\theta+\sin^2\theta)+f^2(\sin^2\theta+\cos^2\theta)$
$=g^2+f^2$

Which remains the same as in given equation.

Ques 05 : Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point (2, -1) and inclined at an angle $tan^{-1} - \frac{4}{3}$

For the 1st transformation, we put $x = x_1 + 2$ and $y = y_1 - 1$ in the given equation, So we obtain-

$11(x_1 + 2)^2 + 24(x_1 + 2)(y_1 - 1) + 4(y_1 - 1)^2 - 20(x_1 + 2) - 40(y_1 - 1) - 5 = 0$

Or, $11x_1^2 + 44x_1 + 44 + 24x_1y_1 - 24x_1y_1 - 24x_1 + 48y_1 - 48 + 4y_1^2 - 8y_1 + 4 - 20x_1 - 40 - 40y_1 + 40 - 5 = 0$

Or, $11x_1^2 + 24x_1y_1 + 4y_1^2 - 5 = 0$ .....(1)

For the 2nd transformation, $\theta = \tan^{-1}\frac{-4}{3}$ or, $tan\theta = -\frac{4}{3}$

$\therefore sin\theta = -\frac{4}{5}$ and $cos\theta = \frac{3}{5}$

Now, let –

$x = x_1 cos\theta - y_1 sin\theta = \frac{3x_1 + 4y_1}{5}$

$y = x_1 sin\theta + y_1 sin\theta = \frac{3y_1 - 4x_1}{5}$

Thus, putting the value of x and y in eq. (1), we get –

$11\left(\frac{3x_1+4y_1}{5}\right)^2 + 24\left(\frac{3x_1+4y_1}{5}\right)\left(\frac{3y_1-4x_1}{5}\right) + 4\left(\frac{3y_1-4x_1}{5}\right)^2 - 5 = 0$

Or, $11(3x_1 + 4y_1)^2 + 24(3x_1 + 4y_1)(3y_1 - 4x_1) + 4(3y_1 - 4x_1)^2 - 25 = 0$

Or, $99x_1^2 + 264x_1y_1 + 176y_1^2 + 216x_1y_1 - 288x_1^2 + 288y_1^2 - 384x_1y_1 + 36y_1^2 - 96x_1y_1 + 64x_1^2 - 125 = 0$

Or, $-125x_1^2 + 500y_1^2 - 125 = 0$

Or, $x_1^2 - 4y_1^2 = -1$

Removing the suffixes -

$x^2 - 4y^2 + 1 = 0$

Ques 06 : Determine the angle through which the axes must be rotated to remove the xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$

To remove the xy term, let -
$x = x_1 cos\theta - y_1 sin\theta$
$y = x_1 sin\theta + y_1 cos\theta$

Putting values of x, y in given equation, we get -
$7(x_1 cos\theta - y_1 sin\theta)^2 - 6\sqrt{3}(x_1 cos\theta - y_1 sin\theta)x_1 sin\theta + y_1 cos\theta\ 13(x_1 sin\theta + y_1 cos\theta)^2 = 16$

Or, $7x_1^2 \cos^2\theta - 14x_1 y_1 cos\theta sin\theta + 7y_1^2 \sin^2\theta - 6\sqrt{3}x_1^2 cos\theta sin\theta - 6\sqrt{3}x_1 y_1 \cos^2\theta + 6\sqrt{3}x_1 y_1 \sin^2\theta + 6\sqrt{3}\, y_1^2 sin\theta cos\theta + 13x_1^2 \sin^2\theta + 26x_1 y_1 sin\theta cos\theta + 13y_1^2 \cos^2\theta = 16$

Or, $7x_1^2 \cos^2\theta + 7y_1^2 \sin^2\theta - 6\sqrt{3}x_1^2 cos\theta sin\theta + 6\sqrt{3}y_1^2 sin\theta cos\theta + 13x_1^2 \sin^2\theta + 13y_1^2 \cos^2\theta + x_1 y_1\left(-14 cos\theta sin\theta - 6\sqrt{3}\cos^2\theta + 6\sqrt{3}\sin^2\theta + 26 sin\theta cos\theta\right) - 16 = 0$

Making the co-efficient of $x_1 y_1$ zero, we get –

$-14 cos\theta sin\theta - 6\sqrt{3}\cos^2\theta + 6\sqrt{3}\sin^2\theta + 26 sin\theta cos\theta = 0$

Or, $12 sin\theta cos\theta - 6\sqrt{3}(\cos^2\theta - \sin^2\theta) = 0$
or, $6.2 in\theta cos\theta - 6\sqrt{3}(\cos^2\theta - \sin^2\theta) = 0$
or, $6 sin2\theta - 6\sqrt{3}(cos2\theta) = 0$
or, $sin2\theta - \sqrt{3} cos2\theta = 0$
or, $sin2\theta = \sqrt{3} cos2\theta$
$\therefore tan2\theta = \sqrt{3}$

So, $2\theta = 60°$ or, $\theta = 30°$

Ques 07 : By transforming to parallel axes through a properly chosen point (h, k) prove that the equation $2x^2 + y^2 - xy - 5x - 4y + 11 = 0$ can be reduced to one containing terms of the 2nd degree only.

Putting $x = x_1 + h$ and $y = y_1 + k$ in the given equation, we obtain –

$2(x_1 + h)^2 + (y_1 + k)^2 - (x_1 + h)(y_1 + k) - 5(x_1 + h) - 4(y_1 + k) + 11 = 0$ .....(1)

Or, $2x_1^2 + 2h^2 + y_1^2 + k^2 - x_1 y_1 - hk - 5h - 4k + 11 + x_1(4h - k - 5) + y_1(2k - h - 4) = 0$

Making the co-efficient of x_1 and y_1 zero-
$4h - k - 5 = 0$
$2k - h - 4 = 0$

Solving these two equations, we get -
$h = 2$ and $k = 3$

Hence, the point is (2, 3) and the educed eq, is $2x^2 - xy + y^2 = 0$

Ques 08 : The direction of axes remaining the same, choose a new origin such that the new co-ordinate of the pair of points whose old co-ordinates are (5,-13) and (-3, 11) may be the forms (h, k) and (-h, -k)

Let the new origin be (α, β), Then -
$5 = h + \alpha$ .....(1)
$-13 = k + \beta$ .....(2)

$-3 = -h + \alpha$ .....(3)
$11 = -k + \beta$ .....(4)

Adding (1) and (3), we get $2\alpha = 2 \quad \therefore \alpha = 1$
Adding (2) and (4), we get $2\beta = -2 \quad \therefore \beta = -1$

So, the new origin is (1, -1)

Ques 09 : Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x, y and xy, both sets of axes being rectangular.

Let, $x = x_1 + h$ and $y = y_1 + k$ in the given eq. Then, we get –

$$17(x_1 + h)^2 + 18(x_1 + h)(y_1 + k) - 7(y_1 + k)^2 - 16(x_1 + h) - 32(y_1 + k) - 18 = 0$$

Or, $x_1(34h + 18k - 16) + y_1(18h - 14k - 32) + 17x_1^2 + 17h^2 + 18x_1y_1 + 18hk - 7y_1^2 - 7k^2 - 16k - 32k - 18 = 0$

Making the co-efficient of x_1 and y_1 zero-
$34h + 18k - 16 = 0$
$18h - 14k - 32 = 0$

Solving these eq., we obtain -
$h = 1$ and $k = -1$

So, the given eq. becomes -
$17x^2 + 18xy - 7y^2 = 10$ .....(1)

Putting $x = x_1 cos\theta - y_1 sin\theta$ and $y = x_1 sin\theta + y_1 cos\theta$ in eq, (1),

$17(x_1 cos\theta - y_1 sin\theta)^2 + 18(x_1 cos\theta - y_1 sin\theta)(x_1 sin\theta + y_1 cos\theta) - 7(x_1 sin\theta + y_1 cos\theta)^2 = 10$ .....(2)

Or, $x_1 y_1(-34 cos\theta sin\theta - 14 sin\theta cos\theta + 18\cos^2\theta - 18\sin^2\theta) + 17x_1\cos^2\theta + 17y_1\sin^2\theta + 18x_1^2 cos\theta sin\theta - 18y_1^2 cos\theta sin\theta - 7x_1^2\sin^2\theta - 7y_1^2\cos^2\theta - 10 = 0$

Making the co-efficient of $x_1 y_1$ zero -
$-34 cos\theta sin\theta - 14 sin\theta cos\theta + 18\cos^2\theta - 18\sin^2\theta = 0$
or, $-48 cos\theta sin\theta + 18(\cos^2\theta - \sin^2\theta) = 0$
or, $18 cos2\theta = 24 sin2\theta$
or, $tan2\theta = \frac{3}{4}$
$\therefore \theta = \tan^{-1}\left(\frac{3}{4}\right)$

So, $cos\theta = \frac{3}{\sqrt{10}}$ and $sin\theta = \frac{1}{\sqrt{10}}$

Putting these values in eq. (2), we obtain –
$2x^2 - y^2 = 1$

Ques 10 : Transform the axes inclined at 30° to the original axes the equation, $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$

Let,
$x = x_1 cos\theta - y_1 sin\theta = \frac{\sqrt{3}x_1 - y_1}{2}$
$y = x_1 sin\theta + y_1 cos\theta = \frac{x_1 + \sqrt{3}y_1}{2}$

So, the given equation becomes -
$\left(\frac{\sqrt{3}x_1 - y_1}{2}\right)^2 + 2\sqrt{3}\left(\frac{\sqrt{3}x_1 - y_1}{2}\right)\left(\frac{x_1 + \sqrt{3}y_1}{2}\right) - \left(\frac{x_1 + \sqrt{3}y_1}{2}\right)^2 = 2a^2$

Or, $(\sqrt{3}x_1 - y_1)^2 + 2\sqrt{3}(\sqrt{3}x_1 - y_1)(x_1 + \sqrt{3}y_1) - (x_1 + \sqrt{3}y_1)^2 = 8a^2$

Or, $3x_1^2 - 2\sqrt{3}x_1y_1 + y_1^2 + 6x_1^2 + 6\sqrt{3}x_1y_1 - 2\sqrt{3}x_1y_1 - 6y_1^2 - x_1^2 - 2\sqrt{3}x_1y_1 - 3y_1^2 = 8a^2$

Or, $8x_1^2 - 8y_1^2 = 8a^2$

$\therefore x_1^2 - y_1^2 = a^2$

Removing the suffixes -
$x^2 - y^2 = a^2$

Ques 11 : The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 = 1$ when referred to rectangular axes through the point (2, 3). Find the inclination of the latter axes to the former.

Given that,
$3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ .....(1)

Putting $x = x_1 + 2$ and $y = y_1 + 3$ in eq. (1), we obtain –

$3(x_1 + 2)^2 + 2(x_1 + 2)(y_1 + 3) + 3(y_1 + 3)^2 - 18(x_1 + 2) - 22(y_1 + 3) + 50 = 0$

Or, $3x_1^2 + 12x_1 + 12 + 2x_1y_1 + 6x_1 + 4y_1 + 12 + 3y_1^2 + 18y_1 + 27 - 18x_1 - 36 - 22y_1 - 66 + 50 = 0$
or, $3x_1^2 + 2x_1y_1 + 3y_1^2 - 1 = 0$

Removing suffixes $3x^2 + 2xy + 3y^2 - 1 = 0$ .....(2)

Now, let the axes be turned through an angle θ ,
$\therefore x = x_1 cos\theta - y_1 sin\theta$
$y = x_1 sin\theta + y_1 cos\theta$

Putting the values of x and y in eq. (2),

$3(x_1 cos\theta - y_1 sin\theta)^2 + 2(x_1 cos\theta - y_1 sin\theta)(x_1 sin\theta + y_1 cos\theta) + 3(x_1 sin\theta + y_1 cos\theta)^2 - 1 = 0$ .....(3)

Making the co-efficient of x_1y_1 zero -
$2(\cos^2\theta - \sin^2\theta) = 0$
or, $cos2\theta = 0$

$\therefore 2\theta = 90°$ or, $\theta = 45°$

Putting the value of theta in (3), we get –
$4x_1^2 + 2y_1^2 = 1$

Removing dashes, $4x^2 + 2y^2 = 1$

Chapter 05 : Pair of Straight Lines

Ques 17 : If the two straight lines represented by $x^2(tan^2\phi + cos^2\phi) - 2xytan\phi + y^2sin^2\phi = 0$ make angles α and β with the axis of x, then show that $tan\alpha - tan\beta = 2$.

Given eq. is,

$x^2(tan^2\phi + cos^2\phi) - 2xytan\phi + y^2sin^2\phi = 0$ .....(1)

Comparing this with $ax^2 + 2hxy + by^2 = 0$, we get -

$a = (tan^2\phi + cos^2\phi)$

$h = -tan\phi$

$b = \sin^2\phi$

Let eq. (1) represent two straight lines $(y - m_{1x})(y - m_2x) = 0$. So,

$$m_1 + m_2 = -\frac{2h}{b} = \frac{2tan\phi}{\sin^2\phi} = \frac{2sin\phi}{\sin^2\phi cos\phi} = \frac{2}{sin\phi cos\phi}$$

And, $$m_1m_2 = \frac{a}{b} = \frac{(tan^2\phi+cos^2\phi)}{\sin^2\phi} = \frac{\frac{\sin^2\phi}{\cos^2\phi}+\cos^2\phi}{\sin^2\phi} = \frac{\sin^2\phi+\cos^4\phi}{\sin^2\phi\cos^2\phi}$$

Now, $tan\alpha - tan\beta = m_1 - m_2$

So, $(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1m_2$

$$= \frac{4}{\sin^2\phi\cos^2\phi} - 4\left(\frac{\sin^2\phi+\cos^4\phi}{\sin^2\phi\cos^2\phi}\right)$$

$$= \frac{4(1-\sin^2\phi-\cos^4\phi)}{\sin^2\phi\cos^2\phi}$$

$$= \frac{4(\cos^2\phi-\cos^4\phi)}{\sin^2\phi\cos^2\phi}$$

$$= \frac{4\cos^2\phi(1-\cos^2\phi)}{\sin^2\phi\cos^2\phi}$$

$$= \frac{4\cos^2\phi\sin^2\phi}{\sin^2\phi\cos^2\phi} = 4$$

Thus, $m_1 - m_2 = tan\alpha - tan\beta = \sqrt{4} = 2$

Ques 19 : If the origin is joined by the two points of intersection of the line $lx + my + n = 0$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ there will be two straight lines, show that they'll be coincident if $a^2l^2 + b^2m^2 = n^2$.

Let the straight line,

$lx + my + n = 0$

or, $n = -(lx + my)$

or, $-\frac{(lx+my)}{n} = 1$ .....(1)

Let eq. (1) intersect the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ .....(2)

at two points P and Q. Join OP and OQ. Their joint eq. will be obtained by (1) and (2) [by making the eq. (2) homogenous by (1)] i.e. :

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Or, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1^2$

Or, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(-\frac{(lx+my)}{n}\right)^2 = \frac{l^2x^2+2lxmy+m^2y^2}{n^2}$

Or, $\frac{x^2}{a^2} - \frac{l^2x^2}{n^2} + \frac{y^2}{b^2} - \frac{m^2y^2}{n^2} - \frac{2lxmy}{n^2} = 0$

Or, $x^2\left(\frac{1}{a^2} - \frac{l^2}{n^2}\right) + y^2\left(\frac{1}{b^2} - \frac{m^2}{n^2}\right) - \frac{2lxmy}{n^2} = 0$

Here, $a = \left(\frac{1}{a^2} - \frac{l^2}{n^2}\right)$, $b = \left(\frac{1}{b^2} - \frac{m^2}{n^2}\right)$ and $h = \frac{lm}{n^2}$

We know that, two lines will be coincident if $h^2 = ab$, so-

$\frac{l^2m^2}{n^4} = \left(\frac{1}{a^2} - \frac{l^2}{n^2}\right)\left(\frac{1}{b^2} - \frac{m^2}{n^2}\right)$

or, $\frac{l^2m^2}{n^4} = \left(\frac{n^2-l^2a^2}{a^2n^2}\right)\left(\frac{n^2-m^2b^2}{b^2n^2}\right)$

or, $a^2b^2l^2m^2 = (n^2 - l^2a^2)(n^2 - m^2b^2)$

or, $a^2b^2l^2m^2 = n^4 - n^2(a^2l^2 + b^2m^2) + a^2b^2l^2m^2$

or, $n^4 = n^2(a^2l^2 + b^2m^2)$

$\therefore n^2 = a^2l^2 + b^2m^2$

Ques 21 : Show that the straight lines represented by $ax^2 + 2\lambda xy - ay^2 = 0$ will be perpendicular to each other.

Comparing the given eq. with the general eq. of 2nd degree , we get –

$a = a,\ \ b = -a,\ \ h = \lambda,\ \ g = f = c = 0$

The given eq. will represent a pair of perpendicular lines if,
$\Delta = 0$ and $a + b = 0$

Here,
$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

And, $a + b = a - a = 0$

Thus, the given eq. represents a pair of straight lines.

Ques 23 : Show that the four straight lines represented by $4xy(x^2 - y^2) - tan\alpha(x^2 + 2xy - y^2)(x^2 - 2xy - y^2) = 0$ will make equal angles with each other.

The given equation is,
$4xy(x^2 - y^2) - tan\alpha(x^2 + 2xy - y^2)(x^2 - 2xy - y^2) = 0$

Putting, $x = rcos\theta$ and $y = rsin\theta$ in (1), we obtain-

$4rcos\theta rsin\theta((rcos\theta)^2 - (rsin\theta)^2) - tan\alpha((rcos\theta)^2 + 2rcos\theta rsin\theta - (rsin\theta)^2)((rcos\theta)^2 - 2rcos\theta y - (rsin\theta)^2) = 0$

or, $4r^4 xos\theta sin\theta(\cos^2\theta - \sin^2\theta) - r^4 tan\alpha(\cos^2\theta + 2cos\theta sin\theta - \sin^2\theta)(\cos^2\theta - 2cos\theta sin\theta - \sin^2\theta) = 0$

or, $2sin2\theta cos2\theta - tan\alpha(cos2\theta + sin2\theta)(cos2\theta - sin2\theta) = 0$

or, $sin4\theta - tan\alpha(\cos^2 2\theta - \sin^2 2\theta) = 0$

or, $sin4\theta = tan\alpha cos4\theta$

or, $tan4\theta = tan\alpha$

or, $4\theta = n\pi + \alpha$

$\therefore \theta = \frac{1}{4}(n\pi + \alpha)$ .....(1)

Putting n=0, 1, 2, 3 in eq. (1) :
$\theta_0 = \frac{1}{4}\alpha$

$\theta_1 = \frac{1}{4}(\pi + \alpha)$

$\theta_2 = \frac{1}{4}(2\pi + \alpha)$

$\theta_3 = \frac{1}{4}(3\pi + \alpha)$

Now,

$\theta_1 - \theta_0 = \frac{\pi}{4}$

$\theta_2 - \theta_1 = \frac{\pi}{4}$

$\theta_3 - \theta_2 = \frac{\pi}{4}$

$\therefore$ The lines represented by the given eq. are equally inclined with each other.

Ques 25 : Show that the four straight lines represented by $4xy - 2y^2 + 6x - 12y - 15 = 0$ and $x^2 + 4xy - 2y^2 = 0$ will form a rhombus through the origin.

The given eq. are -

$4xy - 2y^2 + 6x - 12y - 15 = 0$ .....(1)

$x^2 + 4xy - 2y^2 = 0$ .....(2)

Now,

$4xy - 2y^2 + 6x - 12y - 15 = 0$

or, $x^2 + 2x(2y + 3) - (2y^2 + 12y + 15) = 0$ $\quad [ax^2 + bx + c = 0]$

or, $x = \frac{-(2y+3).2 \pm \sqrt{\{(4y+6)^2 + 4.(2y^2+12y+15)\}}}{2}$

$= \frac{-4y-6 \pm \sqrt{16y^2+48y+36+8y^2+48y+60}}{2}$

$= \frac{-4y-6 \pm \sqrt{24y^2+96y+96}}{2}$

$= \frac{-2(2y+3) \pm 2\sqrt{6}\sqrt{y^2+4y+4}}{2}$

$= -(2y + 3) \pm \sqrt{6}(y + 2)$

Or, $x = -2y - 3 + \sqrt{6}y + 2\sqrt{6} \implies x + (2 - \sqrt{6})y + (3 - 2\sqrt{6}) = 0$ .....(3)

And, $x = -2y - 3 - \sqrt{6}y - 2\sqrt{6} \implies x + (2 + \sqrt{6})y + (3 + 2\sqrt{6}) = 0$ .....(4)

Again,

$x^2 + 4xy - 2y^2 = 0$

or, $x = \frac{-4y \pm \sqrt{16y^2 + 4.2y^2}}{2} = \frac{-4y \pm 2\sqrt{6}y}{2} = -2y \pm \sqrt{6}y$

$\therefore x + (2 - \sqrt{6})y = 0$ .....(5)

and, $x + (2 + \sqrt{6})y = 0$ .....(6)

Solving (3) & (4), we get $y = -2$ & $x = 1$

i.e. the point of intersection of (3) & (4) is $A(1, -2)$

Solving (5) & (6), we get $x = 0$ & $y = 0$

i.e. the point of intersection of (5) & (6) is $C(0, 0)$

Subtracting (6) from (3), we get $y = \frac{3-2\sqrt{6}}{2\sqrt{6}}$

Putting the value of y in (3), we get $x = \frac{6+\sqrt{6}}{2\sqrt{6}}$

i.e. the point of intersection of (3) & (6) is $B(\frac{6+\sqrt{6}}{2\sqrt{6}}, \frac{3-2\sqrt{6}}{2\sqrt{6}})$

Subtracting (5) from (4), we get $y = -\frac{3+2\sqrt{6}}{2\sqrt{6}}$

Putting the value of y in (4), we get $x = \frac{-6+\sqrt{6}}{2\sqrt{6}}$

i.e. the point of intersection of (5) & (4) is $D(\frac{\sqrt{6}-6}{2\sqrt{6}}, -\frac{3+2\sqrt{6}}{2\sqrt{6}})$

Now, $AB = \sqrt{\left\{\left(1 - \frac{6+\sqrt{6}}{2\sqrt{6}}\right)^2 + \left(-2 - \frac{3-2\sqrt{6}}{2\sqrt{6}}\right)^2\right\}} = \sqrt{\frac{75}{24}}$

In this way, $BC = CD = DA = \sqrt{\frac{75}{24}}$

and, $Diagonal\ AC = \sqrt{5}$ $and, BD = \sqrt{\frac{15}{2}}$

Therefore, AB=BC=CD=DA=$\sqrt{\frac{75}{24}}$ and diagonal $BD \neq AC, and\ the\ lines$ (3)& (5)and (4)& (6)$are\ parallel\ to\ each\ other.$

Now, mid point of AC is $\left(\frac{1+0}{2}, -\frac{2}{2}\right) = \left(\frac{1}{2}, -1\right)$

mid point of BD is $\left(\frac{6+\sqrt{6}+\sqrt{6}-6}{2.2\sqrt{6}}, \frac{3-2\sqrt{6}-3-2\sqrt{6}}{2\sqrt{6}}\right) = \left(\frac{1}{2}, -1\right)$

Hence, ABCD is a rhombus.

Ques 27 : Prove that the angles between the straight lines $ax^2 + 2hxy + by^2 + \lambda(x^2 + y^2) = 0$ have the same pair of bisectors. Interpret the case when $\lambda = -(a + b)$.

The eq. of the bisectors of the angle between the given lines is $\frac{x^2-y^2}{a-b} = \frac{xy}{h}$ which is free from λ. Hence all the line-pairs represented by the given eq. have the same bisectors of the angles between them

When $\lambda = -(a + b)$, the given eq. becomes $ax^2 + 2hxy + by^2 - (a + b)(x^2 + y^2) = 0$ or,$bx^2 - 2hxy + ay^2 = 0$. This represents a pair of lines which include the same angle as the pair $x^2 + 2hxy + by^2 = 0$. Also the two pairs have the same bisectors of the angles between them.

Ques 29 : If the eq. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two straight lines, prove that the square of the distance of their point of intersection from the origin is $\frac{c(a+b)^2-f^2-g^2}{ab-h^2}$.

Given eq. is ,
$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ .....(1)
which is the general eq. of 2nd degree.

We know that, the intersection point of the two lines represented by the general eq. of 2nd degree is –

$\left(\frac{hf-bh}{ab-h^2}, \frac{hg-af}{ab-h^2}\right)$ [from theory, chap 05, page 31]

Now, the distance from the intersection (x,y) to the point of origin (0,0) is

$$\begin{aligned}(distance)^2 &= (x-0)^2 + (y-0)^2 \\ &= x^2 + y^2 \\ &= \left(\frac{hf-bh}{ab-h^2}\right)^2 + \left(\frac{hg-af}{ab-h^2}\right)^2 \\ &= \frac{(hf-bh)^2+(hg-af)^2}{(ab-h^2)^2} \\ &= \frac{h^2f^2-2hfbh+b^2g^2+h^2g^2-2hgaf+a^2f^2}{(ab-h^2)^2} \\ &= \frac{\{a(af^2-2fgh)+b(bg^2-2fgh)+h^2(f^2+g^2)\}}{(ab-h^2)^2}\end{aligned}$$

From, $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, we get-
$af^2 - 2fgh = abc - bg^2 - ch^2$
and, $bg^2 - 2fgh = abc - af^2 - ch^2$

So,

$$x^2 + y^2 = \frac{a(abc-bg^2-ch^2)+b(abc-af^2-ch^2)+h^2(f^2+g^2)}{(ab-h^2)^2}$$

$$= \frac{a^2bc+ab^2c-h^2(ac+bc)-ab(g^2+f^2)+h^2(f^2+g^2)}{(ab-h^2)^2}$$

$$= \frac{ab(ac+bc)-h^2(ac+bc)-(g^2+f^2)(ab-h^2)}{(ab-h^2)^2}$$

$$= \frac{(ac+bc)(ab-h^2)-(g^2+f^2)(ab-h^2)}{(ab-h^2)^2}$$

$$= \frac{(ab-h^2)(ac+bc-g^2-f^2)}{(ab-h^2)^2}$$

$$= \frac{c(a+b)-g^2-f^2}{ab-h^2}$$ [Proved]

Ques 31: Show that the four straight lines represented by $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ and $4x^2 - 4xy + y^2 + 4x - 2y - 3 = 0$ will form a Parallelogram.

Same as Ques 25. After getting the lengths of all the sides and diagonals, show that the pair of sides are equal and parallel, and the diagonals are equal, hence a parallelogram.

Ques 33: Find the eq. of the bisectors of the angles between the lines represented by-

(a) $12x^2 + 7xy - 10y^2 + 13x + 45y - 35 = 0$

Here,

$a = 12, \quad b = -10, \quad h = \frac{7}{2}, \quad g = \frac{13}{2}, \quad f = \frac{45}{2}$

So, the point of intersection of the two lines of the given eq. is

$$\left(\frac{hf-bg}{ab-h^2}, \quad \frac{hg-af}{ab-h^2}\right) = \left(-\frac{25}{23}, \frac{43}{23}\right)$$

So, the eq. of bisector of the new origin is –

$$\frac{x_1^2-y_1^2}{a-b} = \frac{x_1y_1}{h}$$

Or, $\frac{x_1^2-y_1^2}{12-(-10)} = \frac{x_1y_1}{\frac{7}{2}}$

Or, $7x_1^2 - 7y_1^2 - 44x_1y_1 = 0$

Transferring to old axes, [Putting $x_1 = x - \left(-\frac{25}{23}\right)$ and $y_1 = y - \frac{43}{23}$]

$$7\left(x + \frac{25}{23}\right)^2 - 44\left(x + \frac{25}{23}\right)\left(y - \frac{43}{23}\right) - 7\left(y - \frac{43}{23}\right)^2 = 0$$

Or, $7(23x + 25)^2 - 44(23x + 25)(23y - 43) - 7(23y - 43)^2 = 0$

(b) $\boldsymbol{2x^2 + 7xy + 6y^2 + 13x + 22y + 20 = 0}$

Here,

$a = 2, \quad b = 6, \quad h = \frac{7}{2}, \quad g = \frac{13}{2}, \quad f = 11$

So, the point of intersection of the two lines of the given eq. is

$$\left(\frac{hf - bg}{ab - h^2}, \qquad \frac{hg - af}{ab - h^2}\right) = (2, -3)$$

So, the eq. of bisector of the new origin is –

$\frac{x_1^2 - y_1^2}{a - b} = \frac{x_1y_1}{h}$

Or, $\frac{x_1^2 - y_1^2}{2 - 6} = \frac{x_1y_1}{\frac{7}{2}}$

Or, $7x_1^2 - 7y_1^2 + 8x_1y_1 = 0$

Transferring to old axes, [Putting $x_1 = x - 2$ and $y_1 = y - (-3)$]

$$7(x - 2)^2 - 44(x - 2)(y + 3) - 7(y + 3)^2 = 0$$

Or, $7x^2 + 8xy - 7y^2 - 4x - 58y - 83 = 0$

Ques 35: Prove that one of the line represented by the equations $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ will bisect the angle between the other two if $(3a + c)^2(bc + 2cd - 3ad) = (b + 3d)^2(bc + 2ab - 3ad)$

Given eq. is -

$ax^3 + bx^2y + cxy^2 + dy^3 = 0$ .....(1)

Let any one of the lines be $y = mx$, which bisects the angle between the other two lines.

Putting the value of y in eq. (1)-

$ax^3 + bx^2mx + cxm^2x^2 + dm^3x^3 = 0$

or, $x^3(dm^3 + cm^2 + bm + a) = 0$

or, $dm^3 + cm^2 + bm + a = 0$ .....(2)

If m_1, m_2 *and* m_3 be the three roots of eq. (2), then -

$m_1 + m_2 + m_3 = -\frac{c}{d}$ $\Rightarrow m_1 + m_2 = -\frac{c}{d} - m_3$ .....(3)

$m_1m_2 + m_2m_3 + m_3m_1 = \frac{b}{d}$

$m_1m_2m_3 = -\frac{a}{d}$ $\Rightarrow m_1m_2 = -\frac{a}{dm_3}$ .....(4)

If $y = m_3x$ bisects the angle between the other lines $y = m_1x$ & $y = m_2x$. Then $m_1 = tan\theta_1$, $m_2 = tan\theta_2$ & $m_3 = tan\theta_3$. So,

$\tan(\theta_1 + \theta_2) = \tan(2\theta_3)$

Or, $\frac{tan\theta_1 + tan\theta_2}{1 - tan\theta_1 tan\theta_2} = \frac{2tan\theta_3}{1 - \tan^2\theta_3}$

Or $\frac{m_1 + m_2}{1 - m_1m_2} = \frac{2m_3}{1 - m_3^2}$

Or, $\frac{-\frac{c}{d} - m_3}{1 + \frac{a}{dm_3}} = \frac{2m_3}{1 - m_3^2}$

Or, $(-c - dm_3)(1 - m_3^2) = 2(m_3d + a)$

Or, $dm_3^3 + cm_3^2 - 3dm_3 - (c + 2a) = 0$ .....(5)

But, since m_3 is a root, $dm_3^3 + cm_3^2 + bm_3 + a = 0$ .....(6)

Subtracting (6) from (5), we obtain -

$-3dm_3 - bm_3 - (c + 2a) - a = 0$

or, $m_3 = \frac{-(c+3a)}{(b+3d)}$

Now, putting the value of m_3 in eq. (6) -

$d\left(\frac{-(c+3a)}{(b+3d)}\right)^3 + c\left(\frac{-(c+3a)}{(b+3d)}\right)^2 + b\left(\frac{-(c+3a)}{(b+3d)}\right) + a = 0$

Or, $-d(c + 3a)^3 + c(c + 3a)^2(b + 3d) - b(c + 3a)(b + 3d)^2 + a(b + 3d)^3 = 0$

Or, $(3a + c)^2(-cd - 3ad + b + 3dc) - (3d + b)^2(bc + 3ab - ab - 3ad) = 0$

Or, $(3a + c)^2(2cd + bc - 3ad) = (3d + b)^2(2ab + bc - 3ad)$ [Proved]

Chapter 08 : System of Circles

Ques 11 : Show that the circles $x^2 + y^2 - 4x - 7y + 6 = 0$, $x^2 + y^2 + 3x - 14y - 1 = 0$ and $3(x^2 + y^2) + 2x - 35y + 4 = 0$ have a common radical axis.

Given that,

$S_1 \equiv x^2 + y^2 - 4x - 7y + 6 = 0$ ..... (1)

$S_2 \equiv x^2 + y^2 - 3x - 14y - 1 = 0$ ..... (2)

$S_3 \equiv 3(x^2 + y^2) + 2x - 35y + 4 = 0$ .....(3)

Now, radical axis of (1) & (2) is,

$S_1 - S_2 = 0$

or, $x^2 + y^2 - 4x - 7y + 6 - (x^2 + y^2 + 3x - 14y - 1) = 0$

or, $-7x + 7y + 7 = 0$

$\therefore x - y - 1 = 0$ ..... (4)

Now, radical axis of (2) & (3) is,

$S_2 - S_3 = 0$

or, $x^2 + y^2 - 3x - 14y - 1 - \left(x^2 + y^2 + \frac{2}{3}x - \frac{35}{3}y + \frac{4}{3}\right) = 0$

or, $3x - \frac{2}{3}x - 14y + \frac{35}{3}y - 1 - \frac{4}{3} = 0$

or, $7x - 7y - 7 = 0$

$\therefore x - y - 1 = 0$ .....(5)

Finally, radical axis of (3) & (1) is,

$S_3 - S_1 = 0$

or, $x^2 + y^2 + \frac{2}{3}x - \frac{35}{3}y + \frac{4}{3} - (x^2 + y^2 - 4x - 7y + 6) = 0$
or, $\frac{2}{3}x - \frac{35}{3}y + \frac{4}{3} + 4x + 7y - 6 = 0$
or, $\frac{2x-35y+4+12x+21y-18}{3} = 0$
or, $14x - 14y - 14 = 0$
$\therefore x - y - 1 = 0$ .....(6)

Now, the radical axes (4), (5) and (6) are same.
Hence, the given circles have a common radical axis.

Ques 12 : Find the general equation of all circles any pair of which have the same radical axis as the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x + 4y = 6$.

Given that,
$S_1 \equiv x^2 + y^2 - 4 = 0$
$S_2 \equiv x^2 + y^2 + 2x + 4y - 6 = 0$

So, to get the general equation -
$S_1 + \lambda S_2 = 0$
or, $x^2 + y^2 - 4 + \lambda(x^2 + y^2 + 2x + 4y - 6) = 0$
or, $x^2 + y^2 - 4 + \lambda x^2 + \lambda y^2 + 2\lambda x + 4\lambda y - 6\lambda = 0$
or, $x^2(1 + \lambda) + y^2(1 + \lambda) + 2\lambda(x + 2y) = 4 + 6\lambda$
$\therefore (1 + \lambda)(x^2 + y^2) + 2\lambda(x + 2y) = 4 + 6\lambda$ (Ans.)

Ques 13 : Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 = 2ax$ and $x^2 + y^2 = 2by$ and having it centre on the line $\frac{x}{a} - \frac{y}{b} = 2$.

Given that,
$x^2 + y^2 - 2ax = 0$ .....(1)
$x^2 + y^2 - 2by = 0$ .....(2)
$\frac{x}{a} - \frac{y}{b} = 2$ .....(3)

Now, equation of the circle passing through the intersection of (1) & (2) is,
$S_1 + \lambda S_2 = 0$
or, $x^2 + y^2 - 2ax + \lambda(x^2 + y^2 - 2by) = 0$
or, $x^2 + y^2 - 2ax + \lambda x^2 + \lambda y^2 - 2\lambda by = 0$
or, $x^2(1 + \lambda) + y^2(1 + \lambda) - 2ax - 2\lambda by = 0$

or, $(1+\lambda)(x^2+y^2)-2ax-2\lambda by=0$

or, $x^2+y^2-\frac{2a}{1+\lambda}x-\frac{2\lambda b}{1+\lambda}y=0$ .....(4) [dividing by $(1+\lambda)$]

Now, the center of (4) is $(\frac{a}{1+\lambda},\frac{\lambda b}{1+\lambda})$. And, this center lies on eq. (3) . So,

$\frac{a}{a(1+\lambda)}-\frac{\lambda b}{b(1+\lambda)}=2$

or, $\frac{1}{1+\lambda}-\frac{\lambda}{1+\lambda}=2$

or, $\frac{1-\lambda}{1+\lambda}=2$

or, $1-\lambda=2+2\lambda$

or, $3\lambda=-1$

$\therefore\ \lambda=-\frac{1}{3}$

Putting the value of λ in (4), we get,

$x^2+y^2-\frac{2a}{1-\frac{1}{3}}x-\frac{2\times-\frac{1}{3}\times b}{1-\frac{1}{3}}y=0$

or, $x^2+y^2-\frac{6ax}{2}+by=0$

or, $2x^2+2y^2-6ax+2by=0$

or, $x^2+y^2-3ax+by=0$

which is the required circle.

Ques 14 : Find the equation of the circle passing through the points of intersection of the line $2x+3y+4=0$ and the circle $x^2+y^2+2x+3y+4=0$ and cutting the circle $x^2+y^2+5x+6y+7=0$ orthogonally.

Given that,

$x^2+y^2+2x+3y+4=0$ .....(1)

$2x+3y+4=0$ .....(2)

$x^2+y^2+5x+6y+7=0$ .....(3)

Now the equation of the circle passing through the point of intersection of (1) & (2) is,

$S_1-\lambda L=0$

or, $x^2+y^2+2x+3y+4-\lambda(2x+3y+4)=0$ .....(4)

or, $x^2+y^2+2x+3y+4-2\lambda x-3\lambda y-4\lambda=0$

or, $x^2+y^2+(2-2\lambda)x+(3-3\lambda)y+4-4\lambda=0$

Now, the center of this eq. is $(-(1-\lambda),-\frac{3-3\lambda}{2})$

As the circles (3) and (4) are orthogonal, So,

$\therefore 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

or, $2\left(-\frac{5}{2}\right)(-1+\lambda) + 2(-3)\left(-\frac{3-3\lambda}{2}\right) = 7 + 4 - 4\lambda$

or, $5 - 5\lambda + 9 - 9\lambda = 11 - 4\lambda$

or, $-10\lambda = 3$

$\therefore \lambda = -\frac{3}{10}$

Putting the value of λ in eq. (4), we get,

$x^2 + y^2 + 2x + 3y + 4 - \frac{3}{10}(2x + 3y + 4) = 0$

or, $\frac{10(x^2+y^2)+20x+30y+40-6x-9y-12}{10} = 0$

or, $10(x^2 + y^2) + 14x + 21y + 28 = 0$

Which is the required equation of the circle.

Ques 15 : Find the co-ordinates of the limiting points of the co-axial system determined by the circles

$x^2 + y^2 - 2x + 8y + 11 = 0$,

and $x^2 + y^2 + 4x + 2y + 5 = 0$

Given circles are,

$x^2 + y^2 - 2x + 8y + 11 = 0$ .....(1)

$x^2 + y^2 + 4x + 2y + 5 = 0$ .....(2)

The radical axis of (1) & (2) is,

$x^2 + y^2 - 2x + 8y + 11 - x^2 - y^2 - 4x - 2y - 5 = 0$

or, $-6x - 6y + 6 = 0$

$\therefore x - y - 1 = 0$

Now, the equation of any circle co-axial with the given circles is,

$S_1 + \lambda(S_1 - S_2) = 0$

or, $x^2 + y^2 - 2x + 8y + 11 + \lambda(x - y - 1) = 0$

or, $x^2 + y^2 - (2 - \lambda)x + (8 - \lambda)y + (11 - \lambda) = 0$ .....(3)

Centre of this circle is $(\frac{2-\lambda}{2}, -\frac{8-\lambda}{2})$ and radius is,

$R = \sqrt{\left(\frac{2-\lambda}{2}\right)^2 + \left(\frac{8-\lambda}{2}\right)^2 - (11 - \lambda)}$

For, limiting point, the radius is zero,

$\left(\frac{2-\lambda}{2}\right)^2 + \left(\frac{8-\lambda}{2}\right)^2 - (11-\lambda) = 0$

or, $4 - 4\lambda + \lambda^2 + 64 - 16\lambda + \lambda^2 - 44 + 4\lambda = 0$

or, $2\lambda^2 - 16\lambda + 24 = 0$

or, $\lambda^2 - 8\lambda + 12 = 0$

or, $(\lambda - 6)(\lambda - 2) = 0$

$\therefore\ \lambda = 6, 2$

Putting the value of λ in the co-ordinates of centre $(\frac{2-\lambda}{2}, -\frac{8-\lambda}{2})$, we get, (-2, -1) and (0,-3).

Hence the limiting points are (-2, -1), (0, -3).

Ques 17 : Circles are drawn through the point (c, 0) touching the circles $x^2 + y^2 = a^2$, Show that the locus of the poles of the axis of x w.r. to these circles is the curve $4a^2(x-c)^2 = (a^2 - c^2)\{a^2 - (c-2x)^2\}y^2$.

Given circle is, $x^2 + y^2 = a^2$ .....(1)

Let the question of the circle be $x^2 + y^2 + 2gx + 2fy + b = 0$ .....(2)

Since, (2) passes through the point (c, 0). So, we get,

$c^2 + 2gc + b = 0$ .....(3)

Centre of (1) is $O_1(0, 0)$ and radius is $r_1 = a$

Centre of (2) is $O_2(-g, -f)$ and radius is $r_2^2 = g^2 + f^2 - b$

Eq. (1) touches (2). Hence, The distance between the centres of (1) & (2) is equal to the sum of the radius of (1) & (2)

$\therefore \sqrt{g^2 + f^2} = a + \sqrt{g^2 + f^2 - b}$

or, $g^2 + f^2 = a^2 + 2a\sqrt{g^2 + f^2 - b} + g^2 + f^2 - b$

or, $-(a^2 - b) = 2a\sqrt{g^2 + f^2 - b}$

or, $(a^2 - b)^2 = 4a^2(g^2 + f^2 - b)$ .....(4)

$\frac{x_1}{y_1}$ be the pole axis of x w.r. to the circle (2) then the polar pf the point (x_1, y_1) w.r. to the circle (2) is

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + b = 0$

or, $xx_1 + g(x + x_1) + fy_1 + b = 0$ [$y = 0$ on x-axis]

or, $x(g + x_1) + (gx_1 + fy_1 + b)$, the sum of two terms will be zero if they are separately zero. Here, $x \neq 0$

$\therefore g + x_1 = 0 \quad \therefore g = -x_1$
and $gx_1 + fy_1 + b = 0$
$\therefore f = \frac{-gx_1 - b}{y_1}$

From (3), we have,

$b = -2gc - c^2 = -2x_1c - c^2$

$\therefore f = \frac{x_1^2 - 2x_{1c} + c^2}{y_1} = \frac{(x_1 - c)^2}{y_1}$

Putting the value of g, f, b in (4), we get,

$(a^2 - 2cx_1 + c^2)^2 = 4a^2(x_1^2 + \frac{(x_1 - c)^4}{y_1^2} - 2cx_1 + c^2)$

or, $(a^2 - 2cx_1 + c^2)^2 = 4a^2x_1^2 + \frac{4a^2(x_1 - c)^4}{y_1^2} - 2.4a^2cx_1 + 4a^2c^2)$

or, $\frac{4a^2(x_1 - c)^4}{y_1^2} = (a^2 - 2cx_1 + c^2)^2 - 4a^2x_1^2 + 8x_1a^2c - 4a^2c^2$

or, $\frac{4a^2(x_1 - c)^4}{y_1^2} = a^4 + 4c^2x_1^2 + c^4 - 4a^2cx_1 - 4c^3x_1 + 2a^2c^2 - 4a^2x_1^2 + 8x_1a^2c - 4a^2c^2$

$= a^4 + 4c^2x_1^2 + c^4 + 4a^2cx_1 - 4x_1c^3 - a^2c^2 - a^2c^2 - 4a^2x_1^2$

$= a^2(a^2 - c^2 + 4cx_1 - 4x_1^2) - c^2(a^2 - c^2 + 4x_1c - 4x_1^2)$

$= (a^2 - c^2)(a^2 - c^2 + 4cx_1 - 4x_1^2)$

$= (a^2 - c^2)(a^2 - (c^2 - 4cx_1 + 4x_1^2)$

$= (a^2 - c^2)\{a^2 - (c - 2x_1)^2\}$

$\therefore 4a^2(x_1 - c)^4 = (a^2 - c^2)\{a^2 - (c - 2x_1)^2\}y_1^2$

The locus of (x_1, y_1) is given by,
$4a^2(x - c)^4 = (a^2 - c^2)\{a^2 - (c - 2x)^2\}y^2$, Proved.

Ques 19 : If the origins be at one of the limiting points of a system of co-axial circles of which $x^2 + y^2 + 2gx + 2fy + c = 0$ is a member, prove that the equation of the system of circles cutting them all orthogonally is $(x^2 + y^2)(g + \mu f) + c(x + \mu y) = 0$

Any circle of the coaxial system of which $x^2 + y^2 + 2gx + 2fy + c = 0$ is a member,
$x^2 + y^2 = 0$ for a limiting point the eqn. of system of circle is,

$x^2 + y^2 + 2gx + 2fy + c + \lambda(x^2 + y^2) = 0$
or, $(1+\lambda)(x^2 + y^2) + 2gx + 2fy + c = 0$ .....(1)

Any circle $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ .....(2) cuts (1) orthogonally if,

$\frac{2gg_1}{1+\lambda} + \frac{2ff_1}{1+\lambda} = c_1 + \frac{c}{1+\lambda}$

or, $2gg_1 + 2ff_1 - c_1(1+\lambda) - c = 0$ for all values of λ
if $c_1 = 0$ and $f_1 = \mu g_1$, then

$2gg_1 + 2f\mu g_1 = c$
or, $2g_1 = \frac{c}{g+f\mu}$

$\therefore c = 2g_1(g + f\mu)$

$2f_1 = 2\mu g_1 = \frac{\mu c}{g+f\mu}$

Now, putting the value of g_1, f_1, c_1 in (2), we get,

$(x^2 + y^2) + x\left(\frac{c}{g+f\mu}\right) + y\left(\frac{\mu c}{g+f\mu}\right) + 0 = 0$

or, $(x^2 + y^2)(g + f\mu) + c(x + \mu y) = 0$

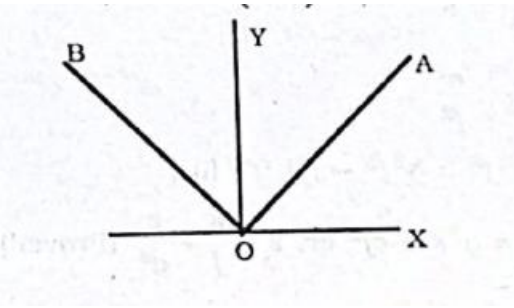
Where μ being a variable parameter. It represents a system of such circles.

Ques 20 : Prove that the limiting points of the system $x^2 + y^2 + 2gx + c + \lambda(x^2 + y^2 + 2fy + k) = 0$ subtend a right angle at the origin if $\frac{c}{g^2} - \frac{k}{f^2} = 2$

The given equation is as,

$x^2 + y^2 + 2gx + c + \lambda(x^2 + y^2 + 2fy + k) = 0$
$(1+\lambda)(x^2 + y^2) + 2gx + 2fy + ky + c = 0$
or, $x^2 + y^2 + \frac{2g}{1+\lambda}x + \frac{2f\lambda}{1+\lambda}y + \frac{c+k\lambda}{1+\lambda} = 0$ .....(1)

Radius of (1) is $R^2 = \left(\frac{g}{1+\lambda}\right)^2 + \left(\frac{f\lambda}{1+\lambda}\right)^2 - \frac{c+k\lambda}{1+\lambda}$



For limiting points radius is zero,

$\therefore g^2 + f^2\lambda^2 - (c + k\lambda)(1 + \lambda) = 0$
or, $g^2 + f^2\lambda^2 - c - c\lambda - k\lambda - k\lambda^2 = 0$
or, $\lambda^2(f^2 - k) - \lambda(k + c) + g^2 - c = 0$ .....(2)

Now, let there be two value of λ , say λ_1, λ_2

$\therefore \lambda_1\lambda_2 = \frac{g^2-c}{f^2-k}$ .....(i) [since product of two roots $\lambda_1\lambda_2$ of (2)]

Centre of (1) is , $O(-\frac{g}{1+\lambda}, \frac{-f\lambda}{1+\lambda})$

$\therefore$ The limiting points are $A(-\frac{g}{1+\lambda_1}, \frac{-f\lambda_1}{1+\lambda_1})$, $B(-\frac{g}{1+\lambda_2}, \frac{-f\lambda_2}{1+\lambda_2})$

OA is perpendicular to OB.
The equation of the line OA is

$$\frac{x+\frac{g}{1+\lambda_1}}{-\frac{g}{1+\lambda_1}} = \frac{y+\frac{f\lambda_1}{1+\lambda_1}}{\frac{f\lambda_1}{1+\lambda_1}}$$

or, $y(1+\lambda_1) = \frac{x(1+\lambda_1)}{g} f\lambda_1$

or, $y = \frac{f\lambda_1 x}{g}$

$\therefore$ slope, $m_1 = \frac{f\lambda_1}{g}$

Similarly, slope of OB is $= \frac{f\lambda_2}{g}$

As OA is perpendicular to OB, So we get

$\frac{f\lambda_1}{g} \times \frac{f\lambda_2}{g} = -1$

or, $\lambda_1\lambda_2 = -\frac{g^2}{f^2}$

or, $g^2k - g^2f^2 = g^2f^2 - cf^2$ [By (i)]

or, $2g^2f^2 = g^2k + cf^2$

or, $\frac{c}{g^2} + \frac{k}{f} = 2$ [Proved]

Ques 21 : Prove that two circles will pass through the two points (0, a) and (0, -a) and touch the straight line $y = mx + c$ will cut orthogonally if $c^2 = a^2(2 + m^2)$

If the circle $x^2 + y^2 + 2gx + 2fy + \lambda = 0$ passes through the points (0, a) and (0, -a), then-

$0 + a^2 + 2g0 + 2fa + \lambda = 0$ or, $a^2 + 2fa + \lambda = 0$

and, $0 + a^2 + 2g0 - 2fa + \lambda = 0$ or, $a^2 - 2fa + \lambda = 0$

Solving these two equations, we get

$f = 0$ and $\lambda = -a^2$. Thus the original equation becomes,

$x^2 + y^2 + 2gx - a^2 = 0$ .....(1)

Centre of eq. (1) is (-g, 0) and radius, $r = \sqrt{g^2 + a^2}$. As the circle touches the line $y = mx + c$, then the perpendicular distance from the centre of the circle to the line is equal to the radius,

Or, $(g^2 + f^2)(1 + m^2) = (c - gm)^2$
or, $g^2 + g^2m^2 + a^2 + a^2m^2 = c^2 - 2cgm + g^2m^2$
or, $g^2 + 2cgm + a^2 + a^2m^2 - c^2 = 0$

If g_1, g_2 be the roots, then $g_1g_2 = a^2 + a^2m^2 - c^2$ and $g_1 + g_2 = 2cm.$

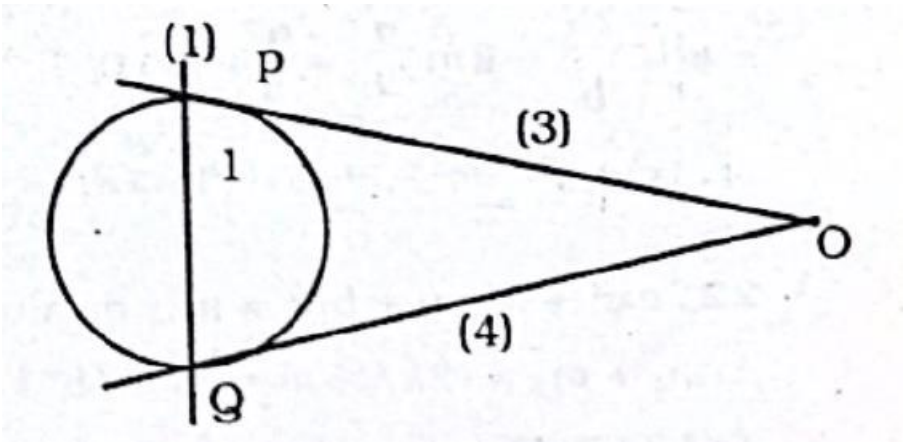
So, we'll get two circles from (1),
$x^2 + y^2 + 2g_1x - a^2 = 0$
$x^2 + y^2 + 2g_2x - a^2 = 0$

These will cut orthogonally if,
$2g_1g_2 + 2f_1f_2 = c_1 + c_2$
or, $2(a^2 + a^2m^2 - c^2) + 0 = -2a^2$
or, $a^2 + a^2m^2 - c^2 + a^2 = 0$
or, $c^2 = 2a^2 + a^2m^2$
or, $c^2 = a^2(2 + m^2)$ [Proved]

Ques 22 : The straight line $lx + my - 1 = 0$ meets the lines $ax^2 + 2hxy + by^2 = 0$ in the points P and Q. Show that the equation of the circle described on PQ as diameter is $S = (x^2 + y^2)(am^2 - 2hlm + bl^2) - 2x(bl - hm) - 2y(am - hl) + a + b = 0$

Given that, $lx + my = 1$ .....(1)
and, $ax^2 + 2hxy + by^2 = 0$ .....(2)

Let, $ax^2 + 2hxy + by^2 = (y - m_1x)(y - m_2x)$
Where, $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$
The lines are $y = m_1x$ .....(3) and $y = m_2x$ .....(4)



Solving (1) and (3), we get,
$x = \frac{1}{l+mm_1}$, and $y = \frac{m_1}{l+mm_1}$

Similarly, solving (1) & (4), we get,
$x = \frac{1}{l+mm_2}$ and $y = \frac{m_2}{l+mm_2}$

So, the co-ordinates are P$(\frac{1}{l+mm_1}, \frac{m_1}{l+mm_1})$ and Q$(\frac{1}{l+mm_2}, \frac{m_2}{l+mm_2})$

Now, equation of the circle described on PQ as diameter is
$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

or, $\left(x-\frac{1}{l+mm_1}\right)\left(x-\frac{1}{l+mm_2}\right)+\left(y-\frac{m_1}{l+mm_1}\right)\left(y-\frac{m_2}{l+mm_2}\right)=0$

or, $\{(l+mm_1)x-1\}\{x(l+mm_2)-1\}+\{y(l+mm_1)-m_1\}\{y(l+mm_2)-m_2\}=0$

or, $x^2(l+mm_1)(l+mm_2)-x(l+mm_1+l+mm_2)+1+y^2(l+mm_1)(l+mm_2)-y\{(l+mm_1)m_2+m_1(l+mm_2)\}+m_1m_2=0$

or, $(x^2+y^2)\{l^2+lm(m_1+m_2)+m^2m_1m_2\}-x\{2l+m(m_1+m_2)\}-y\{l(m_1+m_2)+2mm_1m_2\}+m_1m_2+1=0$

Now, putting the value of m_1+m_2 and m_1m_2 we get,

$\therefore (x^2+y^2)\left\{l^2+ml\left(-\frac{2h}{b}\right)+\frac{m^2a}{b}\right\}-x\left\{2l+m\left(-\frac{2h}{b}\right)\right\}-y\left\{l\frac{-2h}{b}+2m\frac{a}{b}\right\}+\frac{a}{b}+1=0$

Or, $(x^2+y^2)(am^2-2hlm+bl^2)-2x(bl-hm)-2y(am-hl)+a+b=0$ [Proved]